SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Deep Learning COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

Duration: 1 Hour

QUESTIONS: 5 Total Marks:50

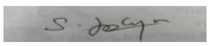
Annexure A

Analytical Problem Solving

- A coin is tossed 10 times and the outcomes are:
  H H T H H H T H H T
  Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
- Out of 50 students, 42 passed an exam.
  Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
- A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
  Find the MLE for the probability that a bulb is defective.
- A biased die is rolled 60 times. The number 6 appears 18 times.
  Estimate the probability of rolling a 6 using MLE.
- Given:
  Input (x) = 4
  Weight (w) = 2
  Actual Output = 10
  Prediction:
  $y = w \times x$
  - Find the predicted output.
  - Calculate the error.
  - If the gradient is 4 and learning rate is 0.01, find the updated weight.

Code of Conduct:

I LOGESH S 192525023 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

solutions:

1. Coin Toss – Maximum Likelihood Estimation (MLE)
Given outcomes:
H H T H H H T H H T

Total tosses, n=10
Number of Heads = 7
The MLE for the probability of Heads is:

p=Number of Heads / Total Tosses

p=7/10=0.7

Answer:
p=0.7

2. Students Passing an Exam
Given:

Total students = 50
Passed = 42
MLE:

p=42 / 50=0.84

Answer:
Probability of passing = 0.84 (84%)

3. Defective Light Bulbs
Given:

Total bulbs = 100

Defective bulbs = 8
MLE:

p= 8 / 100 = 0.08

Answer:
Probability of a bulb being defective = 0.08 (8%)

4. Biased Die
Given:

Total rolls = 60
Number of times 6 appears = 18
MLE:

p= 18 / 60= 0.3

Answer:
Probability of rolling a 6 = 0.3 (30%)

5. Prediction, Error and Weight Update
Given:

Input, x=4
Weight, w=2
Actual Output = 10
Prediction formula:
y= w*x

(i) Predicted Output
y= 2×4 =8

Answer:
Predicted Output = 8

(ii) Calculate the Error

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$
$$= 10 - 8 = 2$$

Answer:
Error = 2

(iii) Updated Weight

Given:

Gradient = 4
Learning Rate = 0.01
Gradient descent update rule:

$$\begin{aligned} w(new) &= w - (\text{Learning Rate} \times \text{Gradient}) \\ &= 2 - (0.01 \times 4) \\ &= 2 - 0.04 \\ &= 1.96 \end{aligned}$$

Answer:
Updated Weight = 1.96